1.1- Propositional Logic

Logic :-

Logic is the study of reasoning. It is the basic on which all the sentences are build. OR Correct and Incorrect reasoning.

Proposition :-

It is a declarative sentence (that is, a sentence that declares fact) that is either true or false, but not both.

Propositional Logic : The area of logic that deals with propositions.

Example :-

1. What time is it? ✗
2. Read this carefully. ✗
3. $x + 1 = 3$ ✗
4. $x + y = z$ ✗
5. Toronto is the capital of Canada. ✓ (T)
6. $1 + 1 = 2$ ✓ (T)
7. $2 + 2 = 3$ ✓ (F)

Sentence park kr samajh ae ke baat true hai ya false agr aisa na ho to wo proposition nahi hai.

Truth Value :-

If any proposition is true then its truth value is denoted by T and if the proposition is false then its truth value is denoted by F.

Example :-

1. 1 is less than 3 → T
2. 14 is odd number. → F

Type Of Proposition :-

a) Simple Proposition :-

The proposition having one subject and one predicate is called a simple proposition.

Example :-

1. This flower is pink.
2. Every even number is divisible by 2.

b) Compound Proposition :-

Two or more simple proposition when combined by various connectivities into a single composite sentence is called compound proposition.

Example :-

1. The earth is round and revolves around the Sun.
2. A triangle is equilateral if its three sides are equal.

Propositional Variables :-

(sentential variables)

Each proposition is represented by a symbol / variable such as $p, q, r, s, \ldots$.

Example :-

- p : "It is raining."
- q : "2 + 2 = 5."

Logical Connectives :-

Logical Connectives are symbols used to combine or modify propositions to form complex statements.

OR

The particular symbols used to join two or more proposition into a single composite form or compound proposition are called logical connectives.

Truth Table: A truth table is a tabular representation of all the combinations of values for inputs and their corresponding outputs.

Symbol	Name	Meaning
$\neg p, p', \bar{p}$	Negation	"Not p" (opposite of p)
$p \wedge q$	Conjuction	"p AND q" (Both must be true)
$p \vee q$	Disjunction	"p OR q" (At least one true)
$p \rightarrow q$	Implication	"IF p THEN q" (Cause - effect)
$p \leftrightarrow q$	Biconditional	"p IF and only if q" (Both same)
$p \oplus q$	Exclusive (XOR)	"Either p OR q (but not both)"
$p \uparrow q$	NAND	"NOT (p AND q)"
$p \downarrow q$	NOR	"NOT (p OR q)"

1) Negation ($\neg p$) → "NOT p"

- Reverse the truth value of a proposition.
- Example:-

p: "Today is Sunday"
$\neg p$: "Today is not Sunday".

2) Conjuction :- $(\wedge) \rightarrow$ AND

Any two proposition can be combined by the word "And" to form a compound proposition said to be the conjuction.

Example :-

1. Karachi is in Pakistan and $2+2=4$. **T**
2. Karachi is in Dubai and $2+2=4$. **F**
3. $2+2=4$ and $2+1=4$ **F**

Truth Table :-

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

3) Disjunction :- $(\vee) \rightarrow$ OR

Any two proposition can be combined by the word "or" to form a compound proposition is said to be the disjunction.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

4) Conditional Statement / Implication :-

Many statements are of the form "If p then q" such statement are said to be the conditional statement and denoted by $p \to q$.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Short trick : jahan first cheez false ho ans true T aega and jahan first T ho wahan agr second bhi T hai to ans T warna ans F.

5) Biconditional Statement / Bi-Implication :-

A statement "p if and only if q" such statement are said to be bi-conditional statement and denoted by $p \leftrightarrow q$ or $p \Leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- "Only true if p and q are same
- False when they are different.

6) Exclusive OR (XOR) :- $\oplus$

"Either p or q but not both".

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

- True when p and q are different.
- False when p and q are same.

Final Summary

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

Quick Shortcut to Remember :-

* Negation ($\neg p$): Flips the truth value.
* AND ($\wedge$): Only true if both are true.
* OR ($\vee$): Only false if both are false.
* Implication ($\to$): Only false if true leads to false.
* Biconditional ($\leftrightarrow$): True if both are the same.
* Exclusive OR ($\oplus$): True if both are different.

Converse, Inverse And Contrapositive Of Conditional Statement :-

1) Converse of Conditional Statement :-

Let $p \to q$ be a conditional statement then $q \to p$ is called its converse.

Example :- If X is a labourer then he is poor.

If X is poor then he is labourer.

2) Inverse of Conditional Statement :-

Let $p \to q$ be a conditional statement then $\sim p \to \sim q$ is called its inverse.

: If Haris read book then he will get knowledge.

If Haris not read book then he will not get knowledge.

3) Contrapositive of Conditional Statemen

Let $p \to q$ be a conditional statement then $\sim q \to \sim p$ is called contrapositive.

Example :-

If x is differentiable then it is continuous.

* If x is not continuous then it is not differentiable.

1.3

Propositional Equivalences

~~Tautology~~ :-

1.3.1 Introduction :-

Propositional equivalences are logical statements that always yield the same truth value regardless of the truth values of their individual components. Denoted as $P \equiv Q$.

Types of Equivalences :-

Tautology :-

is a proposition that is always true.

Example :- $p \vee \neg p$

Contradiction :-

is a proposition that is always false.

Example :- $p \wedge \neg p$

Contingency :-

is a proposition which is neither a tautology nor a contradiction. such as p.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

1.3.2

Logical Equivalences

Compound propositions that have the same truth values in all possible cases are called logically equivalent.

Definition :-

Two propositions P and Q are logically equivalent if and only if $P \leftrightarrow Q$ is a tautology. This is written as:

$$P \equiv Q$$

This means that P and Q have the same truth table.

How to prove logical equivalence ?

There are two main ways to prove that two propositions are equivalent.

a) Using Truth Tables :-

Construct a truth table for both propositions and verify that their final columns match.

b) Using Logical Identities :-

Apply well-known logical equivalences (rules) step by step to transform one expression into another.

Common Logical Equivalences

These are standard identities used to simplify logical expressions.

1. Identity Laws :-

$P \wedge T \equiv P$
$P \vee F \equiv P$

2. Domination Laws :-

$P \vee T \equiv T$
$P \wedge F \equiv F$

3. Idempotent Laws :-

$P \wedge P \equiv P$
$P \vee P \equiv P$

4. Double Negation Law :-

$\neg(\neg P) \equiv P$

5. Commutative Laws :-

$P \vee Q \equiv Q \vee P$
$P \wedge Q \equiv Q \wedge P$

Negation law:-

$p \wedge p' = F$

$p \vee p' = T$

6. Associative Laws :-

$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$

$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$

7. Distributive Laws :-

$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

8. De Morgan's Laws :-

$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

9. Absorption Laws :-

$P \vee (P \wedge Q) \equiv P$

$P \wedge (P \vee Q) \equiv P$

10. Implication Law :-

$P \rightarrow Q \equiv \neg P \vee Q$

11. Biconditional Law :-

$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$

$P \leftrightarrow Q \equiv (\neg P \vee Q) \wedge (\neg Q \vee P)$

$P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q)$

$P \leftrightarrow Q \equiv \neg P \leftrightarrow \neg Q$

Logical Equivalences Involving Conditional Statements :-

$p \to q \equiv \neg p \vee q$

$p \to q \equiv \neg q \to \neg p$

$\neg p \to q \equiv p \vee q$

$p \wedge q \equiv \neg(p \to \neg q)$

$\neg(p \to q) \equiv p \wedge \neg q$

$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$

$(p \to r) \wedge (q \to r) \equiv (p \wedge q) \to r$

$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$

$(p \to r) \vee (q \to r) \equiv (p \vee q) \to r$

Logical Equivalences Involving Biconditional Statements :-

$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Normal Form

Let $A(p_1, p_2, p_3, \cdots p_n)$ be a statement formula then the construction of truth table may not be practical always. So we consider alternate procedure known as reduction to normal form.

i) Disjunction Normal Form :-

A statement form which consist of disjunction between conjuction is called DNF

Example :-

$$(p \wedge q) \vee r$$

$$(p \wedge \neg q) \vee (\neg p \wedge r) \vee (r \wedge \neg q)$$

Q. Obtain the DNF of the form $(p \to q) \wedge (\neg p \wedge q)$

solution

$$p \to q \equiv \neg p \vee q$$

so, $(\neg p \vee q) \wedge (\neg p \wedge q)$

Apply distributive law.

$\Rightarrow (\neg p \wedge \neg p \wedge q) \vee (q \wedge \neg p \wedge q)$

$\Rightarrow (\neg p \wedge q) \vee (q \wedge \neg p)$

ii) Conjuction Normal Form :-

A statement form which consists of conjuction of disjunction is called CNF.

Example :-

i) $p \wedge q$

ii) $(\sim p \vee q) \wedge (\sim p \vee r)$

Q. Obtain CNF of the form $(p \wedge q) \vee (\neg p \wedge q \wedge r)$

solution

$\Rightarrow (p \wedge q) \vee (\neg p \wedge q \wedge r)$

Using distributive law:

$\Rightarrow (p \vee (\neg p \wedge q \wedge r)) \wedge (q \vee (\neg p \wedge q \wedge r))$

$\Rightarrow (p \vee \neg p) \wedge (p \vee q) \wedge (p \vee r) \wedge (q \vee \neg p) \wedge (q \vee q) \wedge (q \vee r)$

$\Rightarrow (p \vee q) \wedge (p \vee r) \wedge (q \vee \neg p) \wedge (q \vee r)$

1.3.5 Satisfiability :-

A propositional logic expression is satisfiable if there is atleast one way to assign truth values to its variables that makes the expression true.

Unsatisfiability :-

A compound proposition is unsatisfiable if there is not even a single True result in its truth table.

Valid :-

A compound proposition is valid when it is a tautology.

Invalid :-

A compound proposition is invalid when it is either a contradiction or contingency.

Important take aways :-

* Tautology is always satisfiable but satisfiable is not always tautology.

Summary :-

Tautology	Contradiction	Contingency
Always TRUE	Always False	TRUE OR FALSE
Satisfiable	Unsatisfiable	Satisfiable
Valid	Invalid	Invalid

Ex # 1.4

1.4.1) Introduction :-

- A proposition is a statement that is either true or false. A predicate is a statement that contains variables and can be true or false depending on those variables.

*) Proposition : "7 is greater than 5". (Always true)
*) Predicate : "x is greater than 5". (this depend on x).

1.4.2) Predicates :- (propositional functions)

A predicate is a statement that contains variables and becomes a proposition when values are assigned to those variables.

Predicates are written in the form :

$$P(x)$$

where P is the predicate, and x is the variable.

Predicates can also have more than one variable.

Example :-

$$R(x,y): x + y = 10$$

This predicate depends on two variables x and y.

Logic And Proofs

Domain of Predicate :-

* domain is denoted by $\mathcal{U}$.
- The domain is the set of all values the variable can take.

- Changing the domain can change the truth of the predicate.

1.4.3) Quantifiers :-

In predicate logic, quantifiers specify how many elements in a domain satisfy a given predicate. There are two main types of quantifiers :

1. Universal Quantifier ($\forall$) :-

The universal quantifier $\forall$ means "for all" or "for every". It states that a predicate is true for all elements in a given domain.

$$\forall x\, P(x)$$

"For all x, $P(x)$" or "For every x, $P(x)$".

Example :-

If $P(x) : x > 0$ and U is the integers, then $\forall x\, P(x)$ is false.
If $P(x) : x > 0$ and U is the positive integers, then $\forall x\, P(x)$ is true.

2. Existential Quantifier ($\exists$):-

The existential quantifier $\exists$ means that there is atleast one element for which a statement is true. It is written as:

$$\exists x\, P(x)$$

"There exists atleast one x such that $P(x)$ is true"

Example:-

a) Statement: "There is a prime number that is even".

- Predicate Form:

$$\exists x (P(x) \wedge E(x))$$

- This means "There exists an x such that x is prime and x is even."
- True because $x = 2$ is the only prime number.

b) Statement: "Some students in the class failed the test".

Predicate Form:

$$\exists x\, (F(x))$$

Existential quantifiers applies when atleast one element in the set satisfies the condition.

3. Uniqueness Quantifier ($\exists!$) :-

The uniqueness quantifier ($\exists!$) expresses that exactly one element satisfies a given predicate. It is written as :

$$\exists! x \, P(x)$$

"There exists exactly one x such that $P(x)$ is true".

Example :-

Statement : "There is exactly one captian of the cricket team."

Predicate form :

$$\exists! x \, C(x)$$

Meaning : There is one and only one person x such that x is the captian.

The existential quantifier ($\exists$) alone means "at least one", but the uniqueness quantifier ($\exists!$) ensures there is exactly one.

$$\exists x [P(x) \wedge \forall y \, (P(y) \rightarrow y = x)]$$

We use this instead of ($\exists!$).

1.4.9) Negating Quantified Expressions

"Every student in your class has taken a course in calculus".

$$\forall x\, P(x)$$

where $P(x)$ is the statement "x has taken a course in calculus" and the domain consists of the student in your class! The negation of this statement is "It is not the case that every student in your class has taken a course in calculus".

$$\exists x \neg P(x)$$

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

1) Negating Quantifiers :-

When negating quantified statements, the quantifiers flips.

Original Statement	Negation
$\forall x P(x)$	$\exists x \neg P(x)$
$\exists x P(x)$	$\forall x \neg P(x)$

"Flip the quantifier and push the not inside".

1.5) Nested Quantifiers

Q What are nested quantifiers?

When more than one quantifier is used in a statement and one appears inside another, it's called a nested quantifiers.

$$\forall x \exists y P(x,y)$$

Order Matters :-

Unlike regular multiplication, changing the order of quantifiers can change the meaning.

$\forall x \exists y P(x,y)$: For every x, there is some y that makes $P(x,y)$ true.

$\exists y \forall x P(x,y)$: There is one y that works for every x.

Example :-

Let $Q(x,y)$: $x + y = 0$.

$\forall x \exists y Q(x,y)$ is true

For every real number x, there exists a real number y.

$\exists y \forall x Q(x,y)$ is false

There exists a real number y such that for every real number x, $x + y = 0$.

Nested with Same Quantifier.

$\forall x \forall y\, P(x,y)$ → "For all x and for all y, $P(x,y)$".

$\exists x \exists y\, P(x,y)$ → "There exist x and y such that $P(x,y)$".

Key Takeaways :-

- Nesting = One quantifier inside another.
- The order matters when quantifiers are different.
- Think of inner quantifier as being dependent on the outer one.

* "Everyone has a best friend".

$$\forall x \exists y\, F(x,y)$$

* "There is a person who is friends with everyone".

$$\exists x \forall y\, F(x,y)$$

1.5.7) Negating Nested Quantifiers

To negate nested quantifiers, just follow this golden rule.

Flip each quantifier.
Push the negation inside.

Original	Negated
$\forall x \forall y\, P(x,y)$	$\exists x \exists y \neg P(x,y)$
$\forall x \exists y\, P(x,y)$	$\exists x \forall y \neg P(x,y)$
$\exists x \forall y\, P(x,y)$	$\forall x \exists y \neg P(x,y)$
$\exists x \exists y\, P(x,y)$	$\forall x \forall y \neg P(x,y)$

Example :-

$$\neg(\forall x \exists y\, P(x,y))$$

$$\equiv \exists x \forall y \neg P(x,y)$$

There exists an x such that for all y, $P(x,y)$ is not true.

$$\neg(\exists x \forall y\, L(x,y))$$

$$\equiv \forall x \exists y \neg L(x,y)$$

For every x, there exists a y such that $L(x,y)$ is false.

Real life Example :-

"Every person has a favourite color".

$$\forall x \, \exists y \, F(x, y)$$

Negate :-

$$\exists x \, \forall y \, \neg F(x, y)$$

"There is some person who has no favourite color".